

SQL Joins Assignment – LEFT JOIN, RIGHT JOIN, FULL JOIN

This exercise focuses on SQL joins, especially identifying unmatched rows between tables using LEFT JOIN, RIGHT JOIN, and FULL OUTER JOIN. The dataset models a simple online learning platform.

Database Schema

Tables:

1. students
2. courses
3. enrollments
4. instructors

students		
student_id	student_name	email
1	Alice Johnson	alice@email.com
2	Bob Smith	bob@email.com
3	Charlie Brown	charlie@email.com
4	Diana Prince	diana@email.com
5	Ethan Hunt	ethan@email.com

courses		
course_id	course_name	instructor_id
101	SQL Basics	1
102	Python Fundamentals	2
103	Data Analytics	NULL
104	Cloud Computing	3
105	Machine Learning	NULL

Assignment Questions

1. Display all students and the courses they are enrolled in. Include students who are not enrolled in any course.
2. Find all courses that currently have no students enrolled.
3. Display all instructors and the courses they teach, including instructors who are not assigned to any course.
4. Find all courses that do not have an instructor assigned.
5. Display all students and enrollment information using a RIGHT JOIN.
6. Find students who are not enrolled in any course.
7. Use a FULL OUTER JOIN to display all students and enrollments, including unmatched rows from both tables.

8. Find all courses that have never appeared in the enrollments table.
9. Display all instructors and courses using a FULL OUTER JOIN and identify unmatched rows.
10. Create a report showing: student name, course name, and instructor name. Include rows even if course or instructor information is missing.

Bonus Challenge:

Write a query that lists every student and every course, even if there is no enrollment relationship between them.